**Calculus I Lesson 46**

1) $f(x) = 2\sin^{-1}(x-12)$

Find $f'(x)$.

Hint: Chain Rule: $D_x\left(\sin^{-1} u\right) = \dfrac{1}{\sqrt{1-u^2}} \cdot \dfrac{du}{dx}$

a) Let $u = x - 12$. $\dfrac{du}{dx} = \underline{\quad ? \quad}$

b) $f'(x) = D_x\left(2\sin^{-1} u\right) = 2 \cdot D_x\left(\sin^{-1} u\right) = \underline{\quad ? \quad}$

2) $f(x) = 13\cos^{-1}\left(\frac{x}{12}\right)$

Find $f'(x)$.

Chain Rule: $D_x\left(\cos^{-1} u\right) = \frac{-1}{\sqrt{1-u^2}} \cdot \frac{du}{dx}$

a) Let $u = \frac{x}{12}$. $\frac{du}{dx} = \underline{\quad ? \quad}$

b) $f'(x) = D_x\left(13\cos^{-1}\left(\frac{x}{12}\right)\right) = 13 \cdot D_x\left(\sin^{-1} u\right) = \underline{\quad ? \quad}$

3) $f(x) = \tan^{-1}\left(e^{4x}\right)$

Find $f'(x)$.

Chain Rule: $D_x\left(\tan^{-1} u\right) = \dfrac{1}{1+u^2} \cdot \dfrac{du}{dx}$

a) Let $u = e^{4x}$. $\dfrac{du}{dx} = $ ___?___

b) $f'(x) = D_x\left(\tan^{-1}\left(e^{4x}\right)\right) = D_x\left(\tan^{-1}(u)\right) = $ ___?___

4) $f(x) = \dfrac{\sin^{-1}(7x)}{x}$

Find $f'(x)$.

Quotient Rule for Derivative: $D_x\left(\dfrac{u}{v}\right) = u \cdot D_x(v) + v \cdot D_x(u)$

Chain Rule: $D_x\left(\sin^{-1} u\right) = \dfrac{1}{\sqrt{1-u^2}} \cdot \dfrac{du}{dx}$

a) $D_x\left(\sin^{-1}(7x)\right) = \underline{\quad ? \quad}$

b) $f'(x) = \dfrac{x \cdot D_x\left(\sin^{-1}(7x)\right) - \sin^{-1}(7x) \cdot D_x(x)}{x^2} = \underline{\quad ? \quad}$

5) $f(x) = \sin\left(\cos^{-1} x\right)$

Find $f'(x)$.

Let $u = \cos^{-1} x$.

a) $\dfrac{du}{dx} = \underline{\quad ? \quad}$

b) $f'(x) = D_x\left(\sin\left(\cos^{-1} x\right)\right) = D_x\left(\sin(u)\right) = \cos u \cdot \dfrac{du}{dx} = \underline{\quad ? \quad}$

6) $f(x) = 12\sin^{-1}x$

Find equation of the tangent line at $(0,0)$

a) Find $f'(x) =$ ___?___

b) Slope of tangent line $= f'(0) =$ ___?___

c) Equation of tangent line is = ___?___

Hint: $y - y_1 = m(x - x_1)$

7) $f(x) = \tan^{-1}\left(\frac{x}{4}\right)$

Find equation of the tangent line at $(0,0)$

a) Find $f'(x) = \underline{\quad ? \quad}$

b) Slope of tangent line $= f'(0) = \underline{\quad ? \quad}$

c) Equation of tangent line is $= \underline{\quad ? \quad}$

Hint: $y - y_1 = m(x - x_1)$